Projet Réseau d'Entreprise

Journal Libération – VLANs, Routage, SSH, NAT

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Cisco Packet Tracer

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Introduction

- Création d'un réseau d'entreprise pour le journal **Libération**
- Objectifs :
  - Séparer les services (Rédaction, IT, Serveurs)
  - Assurer la sécurité (SSH, NAT)
  - Simuler un accès Internet
- Outil utilisé : **Cisco Packet Tracer**

Architecture réseau

Équipements :

- SW-HQ-L3 (switch principal)
- SW-HQ-L2-1 – Rédaction
- SW-HQ-L2-2 – IT
- Routeur 2911 (NAT)
- Routeur FAI 2811
- PC-Internet

Rôles :

- Routage inter-VLAN
- Accès Internet simulé
- Tests externes

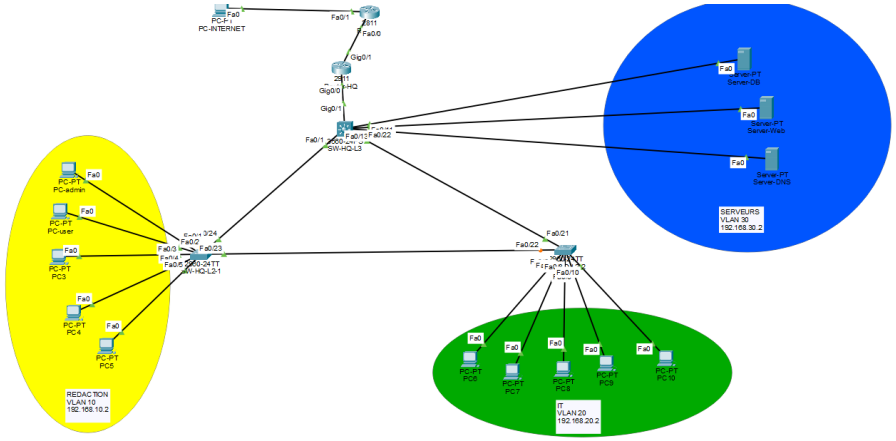

Figure – topologie complète

VLANs et plan d'adressage

VLAN	Nom	Réseau	Passerelle
10	RÉDACTION	192.168.10.0/24	192.168.10.1
20	IT	192.168.20.0/24	192.168.20.1
30	SERVEURS	192.168.30.0/24	192.168.30.1

```
SW-HQ-L3>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   REDACTION                        active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
20   IT                               active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10
30   SERVEURS                         active    Fa0/11, Fa0/12, Fa0/13
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Figure – show vlan brief sur SW-HQ-L3

Attribution des ports par switch

- **SW-HQ-L2-1 (Rédaction)** : Fa0/1-5 → VLAN 10
- **SW-HQ-L2-2 (IT)** : Fa0/6-10 → VLAN 20
- **SW-HQ-L3** : Fa0/11-13 → VLAN 30 (serveurs)

Routage inter-VLAN sur SW-HQ-L3

```
interface vlan 10
 ip address 192.168.10.1 255.255.255.0
 no shutdown
interface vlan 20
 ip address 192.168.20.1 255.255.255.0
 no shutdown
interface vlan 30
 ip address 192.168.30.1 255.255.255.0
 no shutdown
ip routing
```

```
Vlan1                  192.168.0.2     YES manual up                    up
Vlan10                 192.168.10.1    YES manual up                    up
Vlan20                 192.168.20.1    YES manual up                    up
Vlan30                 192.168.30.1    YES manual up                    up
```

Figure – show ip interface brief

```
SW-HQ-L3>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.0.1 to network 0.0.0.0

C    192.168.0.0/24 is directly connected, Vlan1
C    192.168.10.0/24 is directly connected, Vlan10
C    192.168.20.0/24 is directly connected, Vlan20
C    192.168.30.0/24 is directly connected, Vlan30
S*   0.0.0.0/0 [1/0] via 192.168.0.1
```

Figure – show ip route

Configuration des trunks

```
interface fastEthernet 0/23
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
```

```
SW-HQ-L3>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/22      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       10,20,30
Fa0/22      10,20,30

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30
Fa0/22      10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30
Fa0/22      10,20,30
```

Figure – show interfaces trunk

Serveurs (VLAN 30)

Serveur	IP	Rôle
Web	192.168.30.10	HTTP, page Libération
DNS	192.168.30.11	www.liberation.fr → 192.168.30.10
DB	192.168.30.12	Page simulation base de données

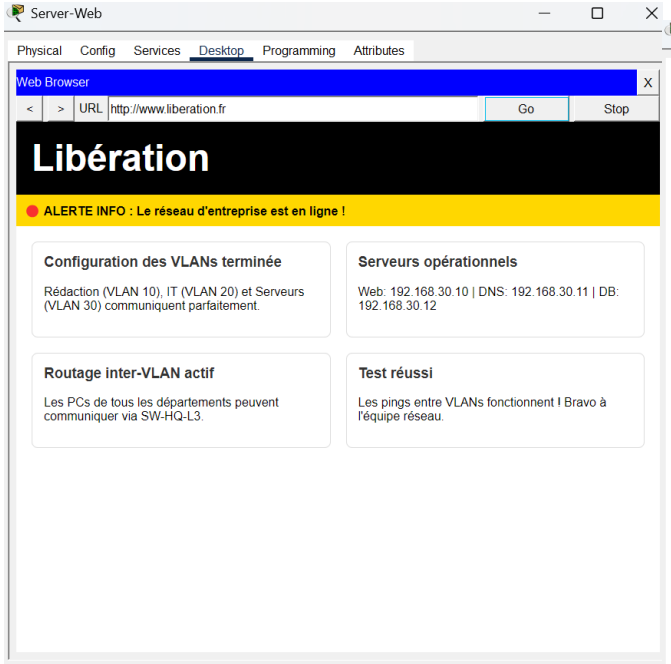

Figure – show ip interface brief

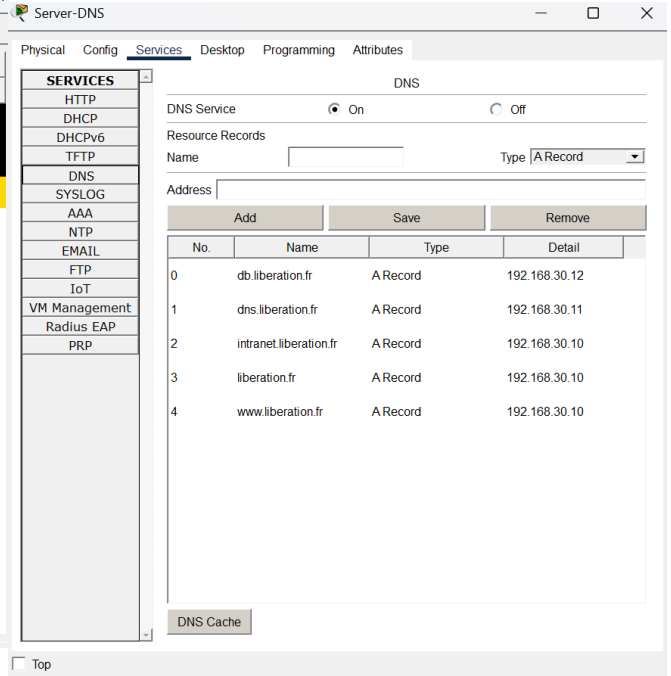

Figure – show ip route

Configuration SSH sur les switches

```
hostname SW-HQ-L3
ip domain-name liberation.local
crypto key generate rsa
username admin secret cisco
line vty 0 15
 transport input ssh
 login local
enable secret cisco
```

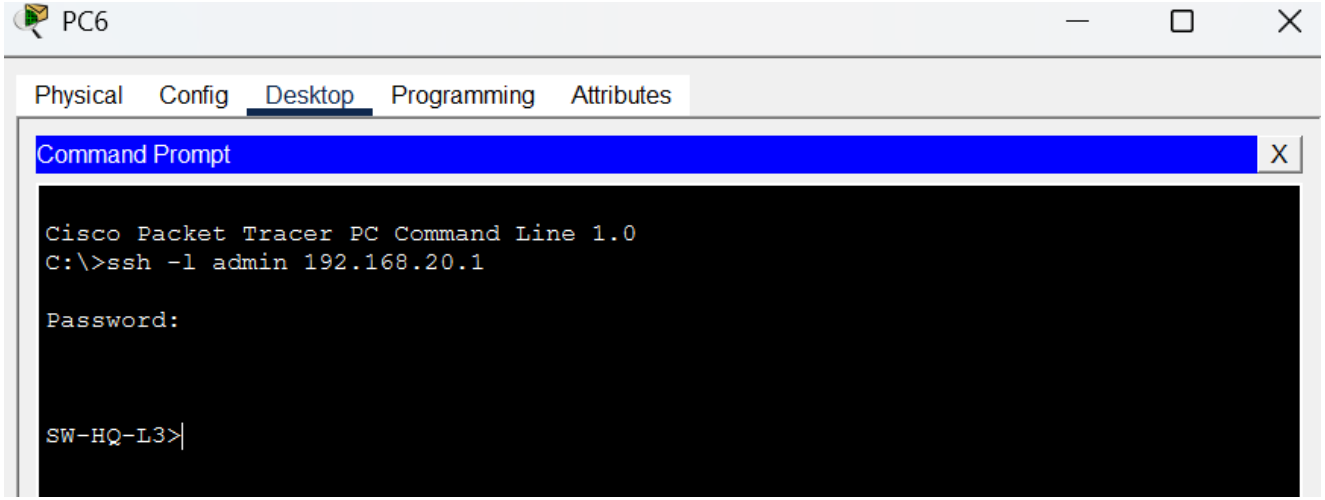

Figure – Connexion SSH depuis un PC IT

Routeur FAI (2811) – Simulation Internet

```
interface fastEthernet 0/0
 ip address 203.0.113.1 255.255.255.0
interface fastEthernet 0/1
 ip address 192.168.100.1 255.255.255.0
ip route 192.168.10.0 255.255.255.0 203.0.113.2
ip route 192.168.20.0 255.255.255.0 203.0.113.2
ip route 192.168.30.0 255.255.255.0 203.0.113.2
```

```
FAI>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    192.168.0.0/24 [1/0] via 203.0.113.2
S    192.168.10.0/24 [1/0] via 203.0.113.2
S    192.168.20.0/24 [1/0] via 203.0.113.2
S    192.168.30.0/24 [1/0] via 203.0.113.2
     192.168.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.100.0/24 is directly connected, FastEthernet0/1
L       192.168.100.1/32 is directly connected, FastEthernet0/1
     203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/24 is directly connected, FastEthernet0/0
L       203.0.113.1/32 is directly connected, FastEthernet0/0
```

Figure – show ip route sur routeur FAI

Configuration NAT (PAT)

```
interface g0/0
 ip address 192.168.0.1 255.255.255.0
 ip nat inside
interface g0/1
 ip address 203.0.113.2 255.255.255.0
 ip nat outside
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
access-list 1 permit 192.168.30.0 0.0.0.255
ip nat inside source list 1 interface g0/1 overload
```

```
Router>show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 0  Misses: 0
Expired translations: 0
Dynamic mappings:
```

Figure – show ip nat statistics

Tests de validation

- **Ping VLAN 10 → passerelle** : 192.168.10.1
- **Ping VLAN 20 → passerelle** : 192.168.20.1
- **Ping VLAN 10 → VLAN 20** : 192.168.20.10
- **Accès Web** : www.liberation.fr
- **DNS** : nslookup www.liberation.fr → 192.168.30.10
- **SSH** : ssh -l admin 192.168.20.1
- **NAT** : ping 203.0.113.1

```
:\>ping 203.0.113.1

inging 203.0.113.1 with 32 bytes of data:

eply from 203.0.113.1: bytes=32 time<1ms TTL=253
eply from 203.0.113.1: bytes=32 time<1ms TTL=253
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eply from 203.0.113.1: bytes=32 time<1ms TTL=253

ing statistics for 203.0.113.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
pproximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure – ping 203.0.113.1 depuis VLAN 10

```
C:\>ping 203.0.113.1

Pinging 203.0.113.1 with 32 bytes of data:

Reply from 203.0.113.1: bytes=32 time<1ms TTL=253
Reply from 203.0.113.1: bytes=32 time<1ms TTL=253
Reply from 203.0.113.1: bytes=32 time<1ms TTL=253
Reply from 203.0.113.1: bytes=32 time<1ms TTL=253

Ping statistics for 203.0.113.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure – ping 203.0.113.1 depuis VLAN 20

Conclusion

Résultats

- Réseau segmenté (VLANs 10, 20, 30)
- Serveurs Web, DNS, DB fonctionnels
- Administration sécurisée (SSH)
- Accès Internet via NAT
- Tests validés